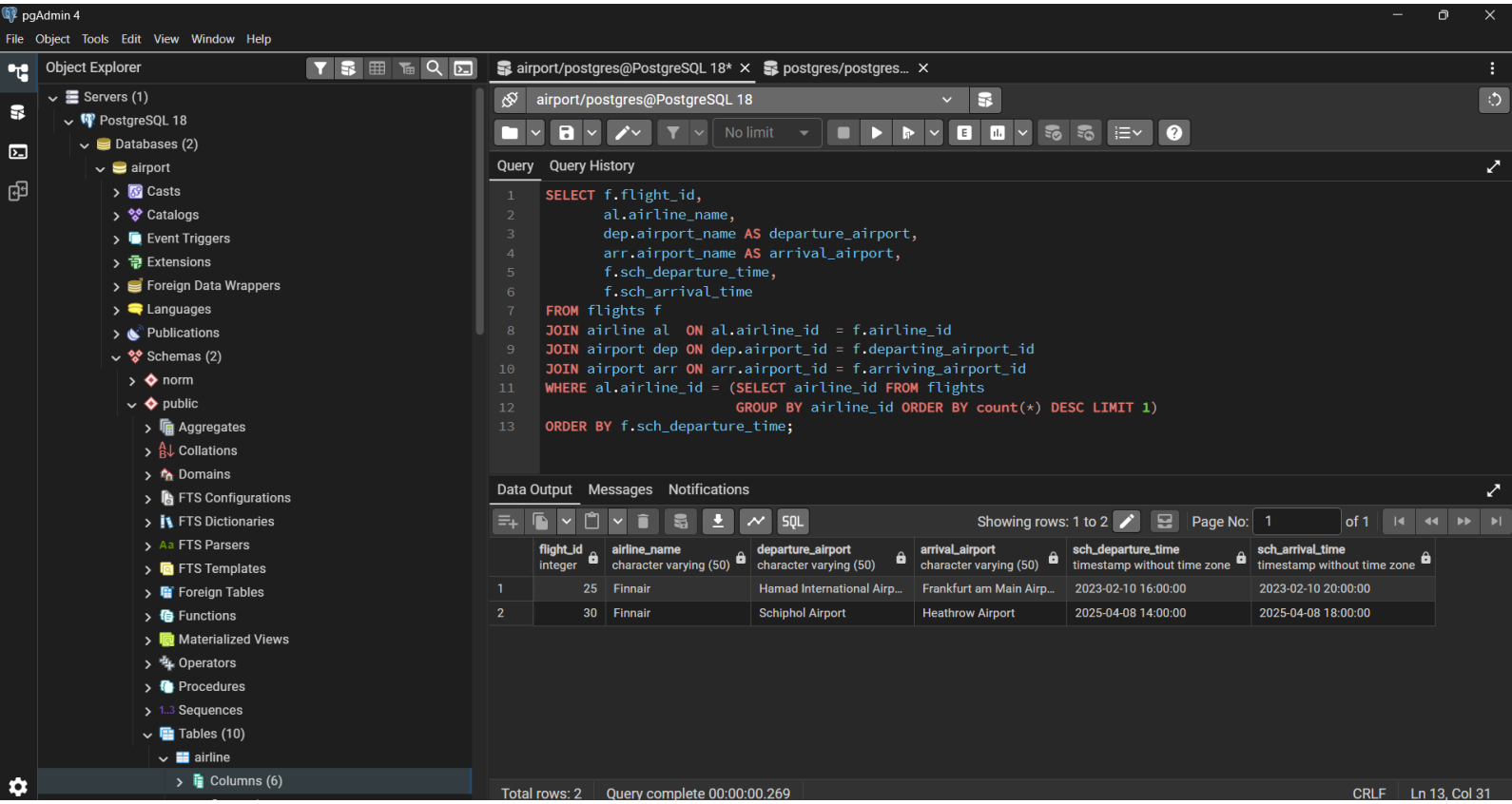
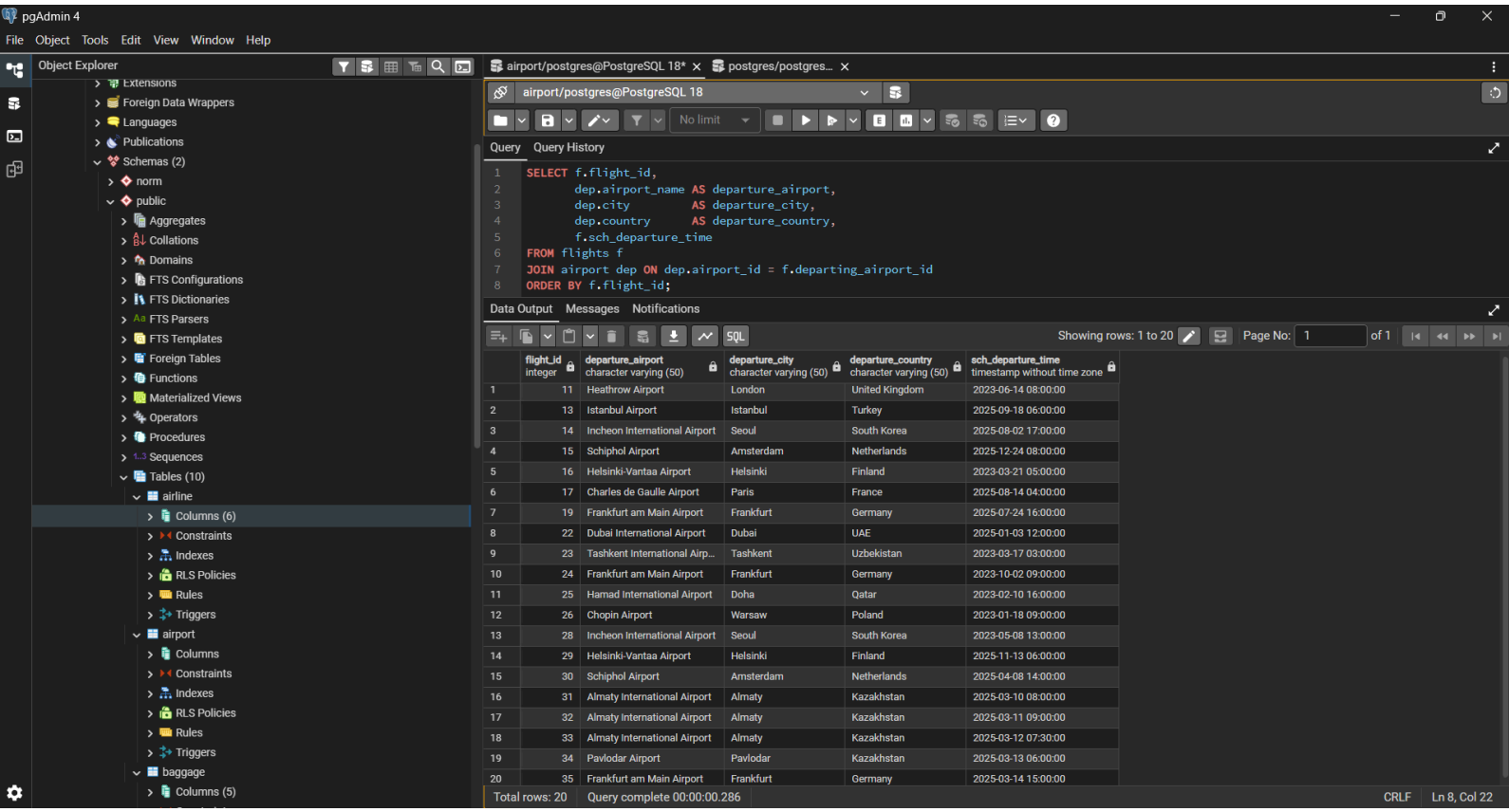
Laboratory work 6

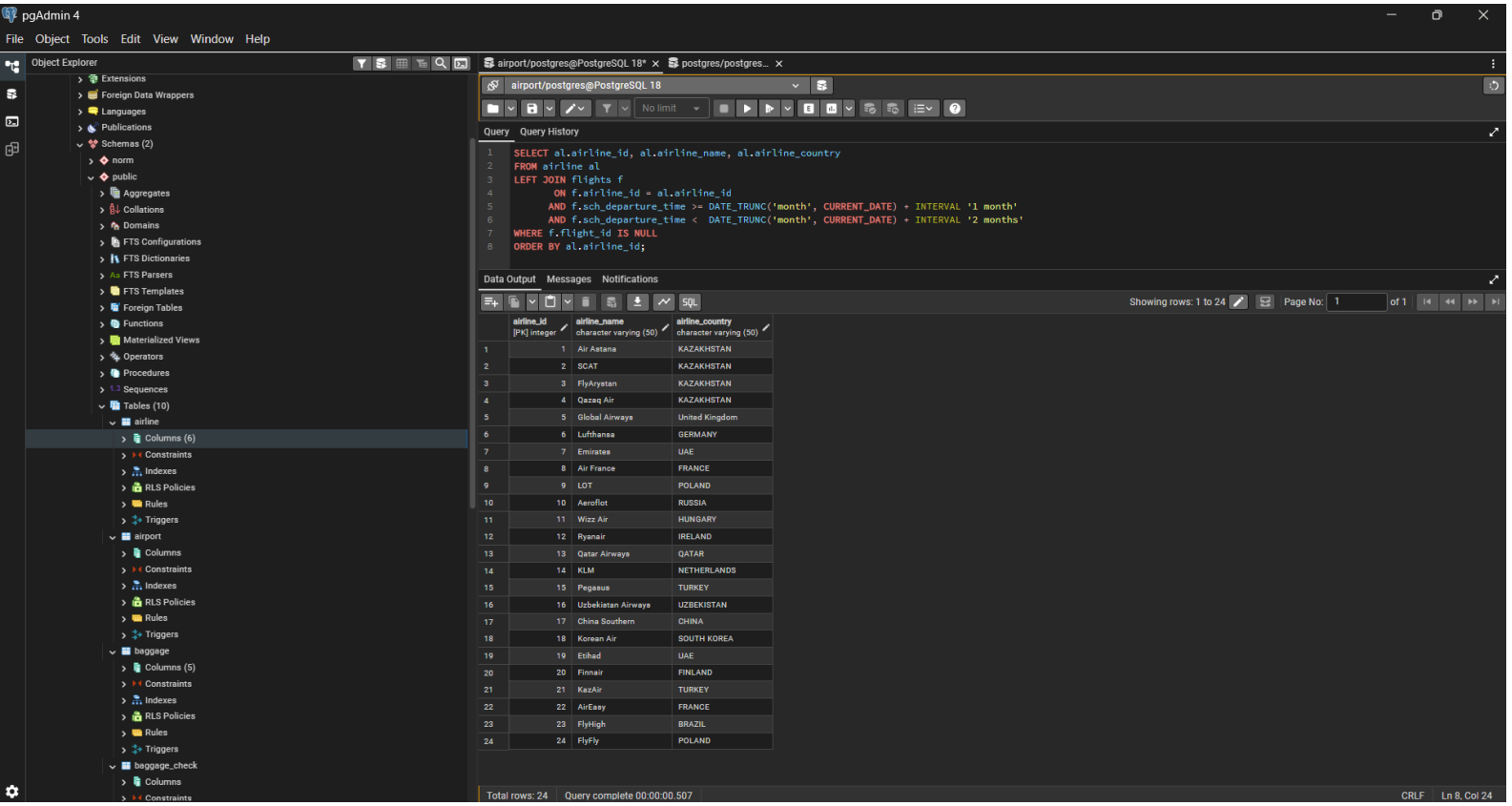
Task 1



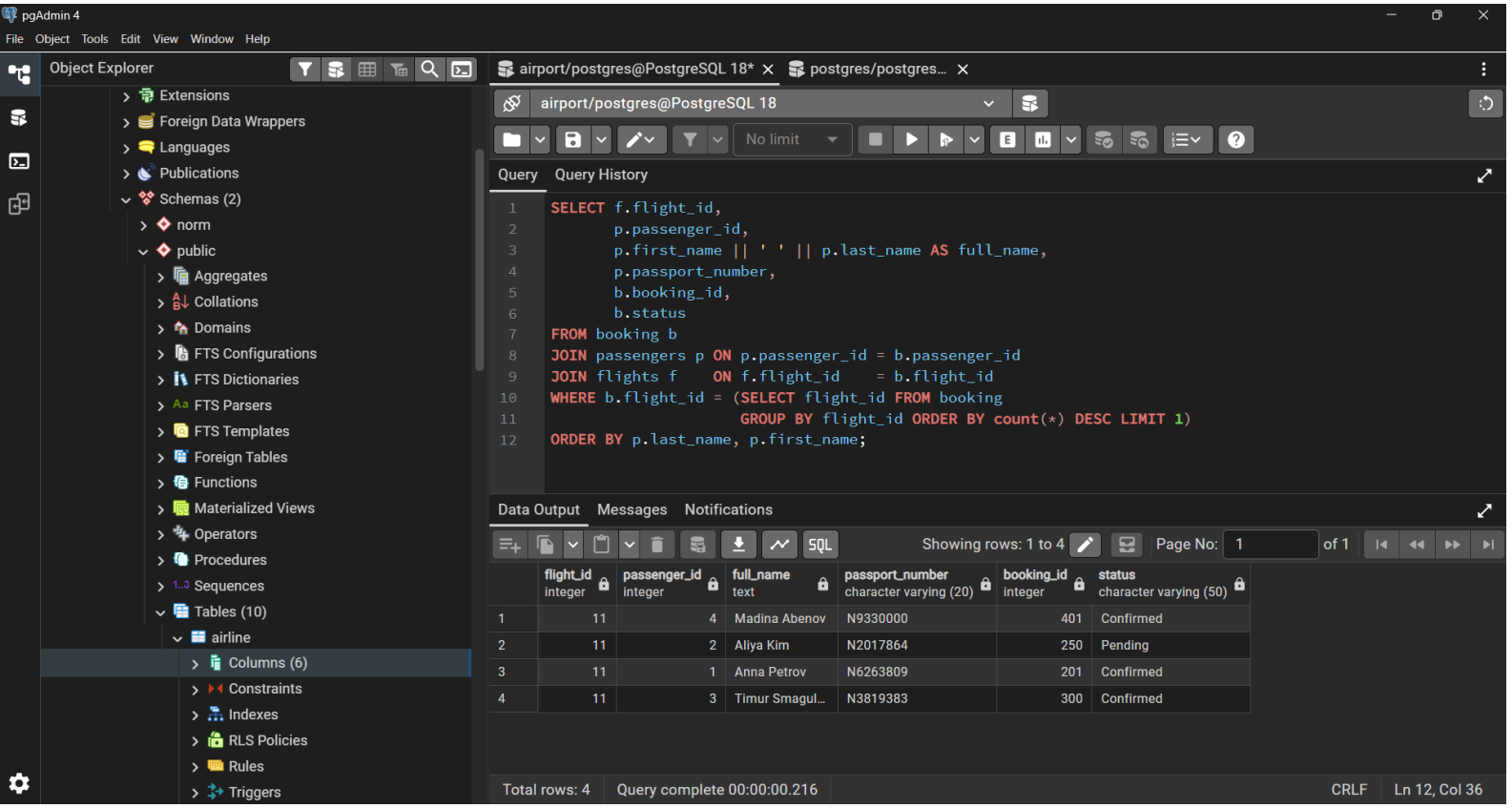
Task 2



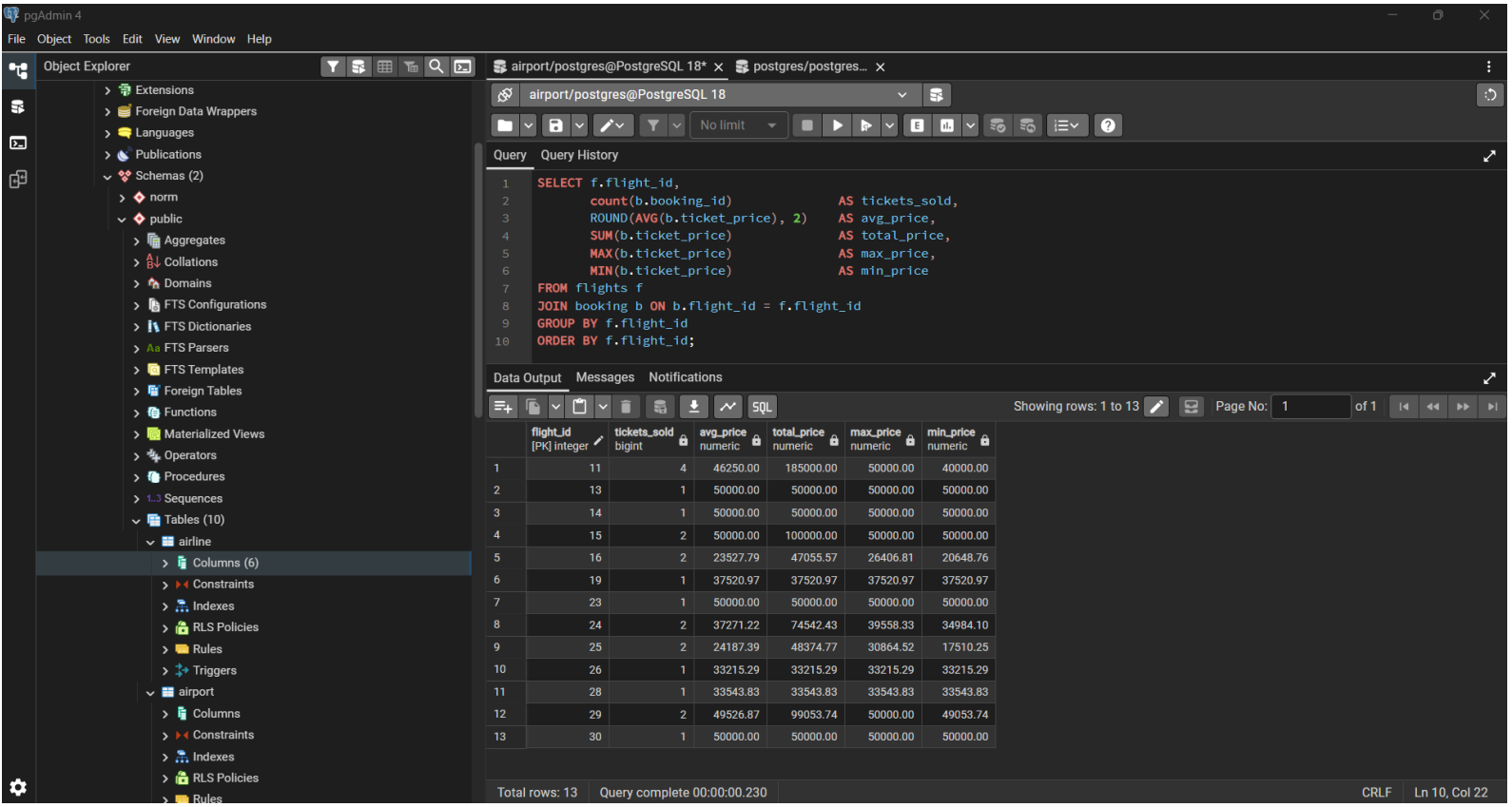
Task 3


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport/postgres@PostgreSQL 18*
postgres/postgres...
airport/postgres@PostgreSQL 18
No limit
Query Query History
SELECT al.airline_id, al.airline_name, al.airline_country
FROM airline al
LEFT JOIN flights f
ON f.airline_id = al.airline_id
AND f.sch_departure_time >= DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '1 month'
AND f.sch_departure_time < DATE_TRUNC('month', CURRENT_DATE) + INTERVAL '2 months'
WHERE f.flight_id IS NULL
ORDER BY al.airline_id;
Data Output Messages Notifications
Showing rows: 1 to 24
Page No: 1 of 1
airline_id [PK] integer
airline_name character varying (50)
airline_country character varying (50)
1 1 Air Astana KAZAKHSTAN
2 2 SCAT KAZAKHSTAN
3 3 FlyArystan KAZAKHSTAN
4 4 Qazaq Air KAZAKHSTAN
5 5 Global Airways United Kingdom
6 6 Lufthansa GERMANY
7 7 Emirates UAE
8 8 Air France FRANCE
9 9 LOT POLAND
10 10 Aeroflot RUSSIA
11 11 Wizz Air HUNGARY
12 12 Ryanair IRELAND
13 13 Qatar Airways QATAR
14 14 KLM NETHERLANDS
15 15 Pegasus TURKEY
16 16 Uzbekistan Airways UZBEKISTAN
17 17 China Southern CHINA
18 18 Korean Air SOUTH KOREA
19 19 Etihad UAE
20 20 Finnair FINLAND
21 21 KazAir TURKEY
22 22 AirEasy FRANCE
23 23 FlyHigh BRAZIL
24 24 FlyFly POLAND
Total rows: 24
Query complete 00:00:00.507
CRLF
Ln 8, Col 24

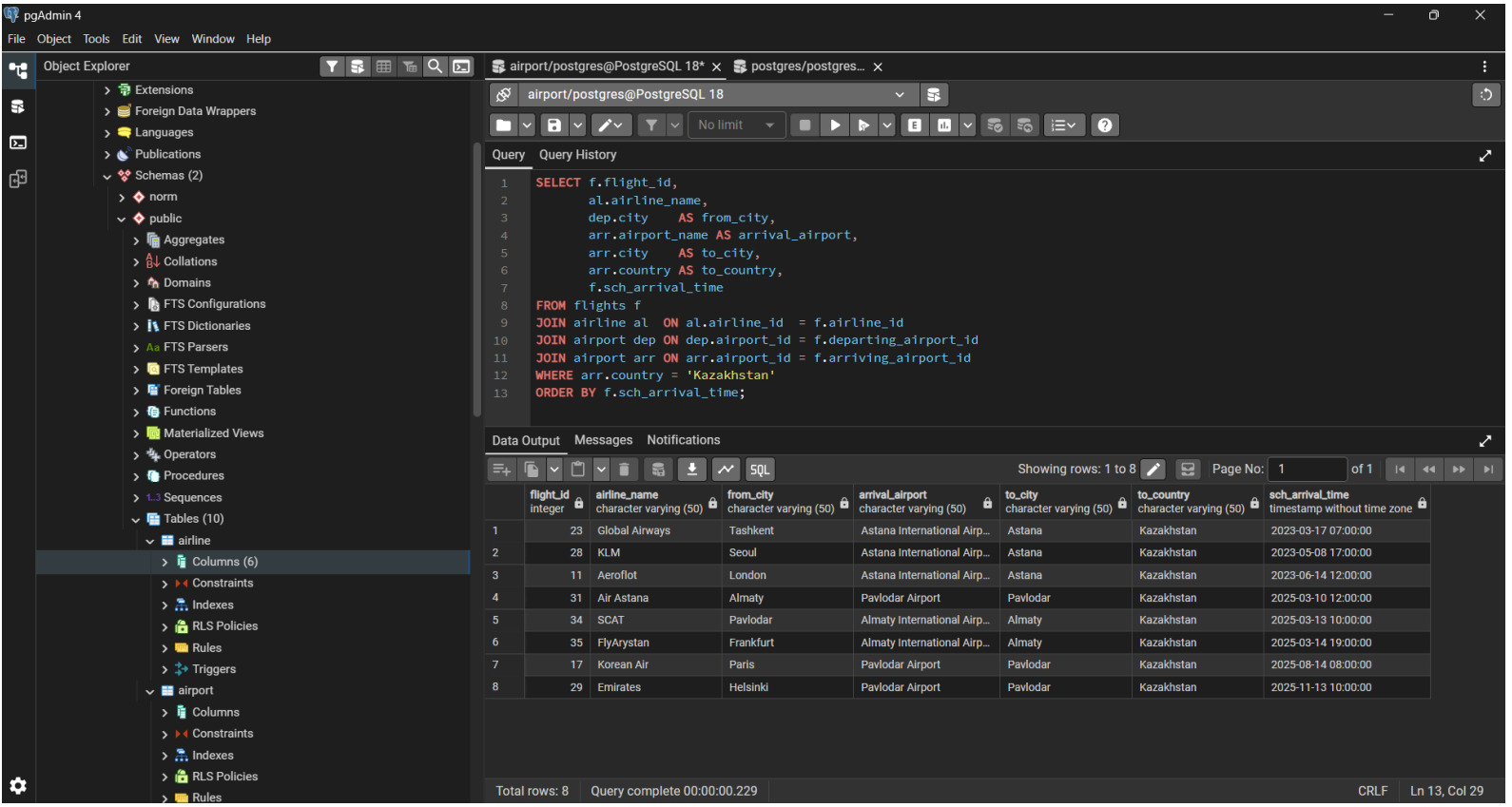
Task 4


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport/postgres@PostgreSQL 18*
postgres/postgres...
airport/postgres@PostgreSQL 18
No limit
Query Query History
SELECT f.flight_id,
p.passenger_id,
p.first_name || ' ' || p.last_name AS full_name,
p.passport_number,
b.booking_id,
b.status
FROM booking b
JOIN passengers p ON p.passenger_id = b.passenger_id
JOIN flights f ON f.flight_id = b.flight_id
WHERE b.flight_id = (SELECT flight_id FROM booking
GROUP BY flight_id ORDER BY count(*) DESC LIMIT 1)
ORDER BY p.last_name, p.first_name;
Data Output Messages Notifications
Showing rows: 1 to 4
Page No: 1 of 1
flight_id integer
passenger_id integer
full_name text
passport_number character varying (20)
booking_id integer
status character varying (50)
1 11 4 Madina Abenov N9330000 401 Confirmed
2 11 2 Aliya Kim N2017864 250 Pending
3 11 1 Anna Petrov N6263809 201 Confirmed
4 11 3 Timur Smagul... N3819383 300 Confirmed
Total rows: 4
Query complete 00:00:00.216
CRLF
Ln 12, Col 36

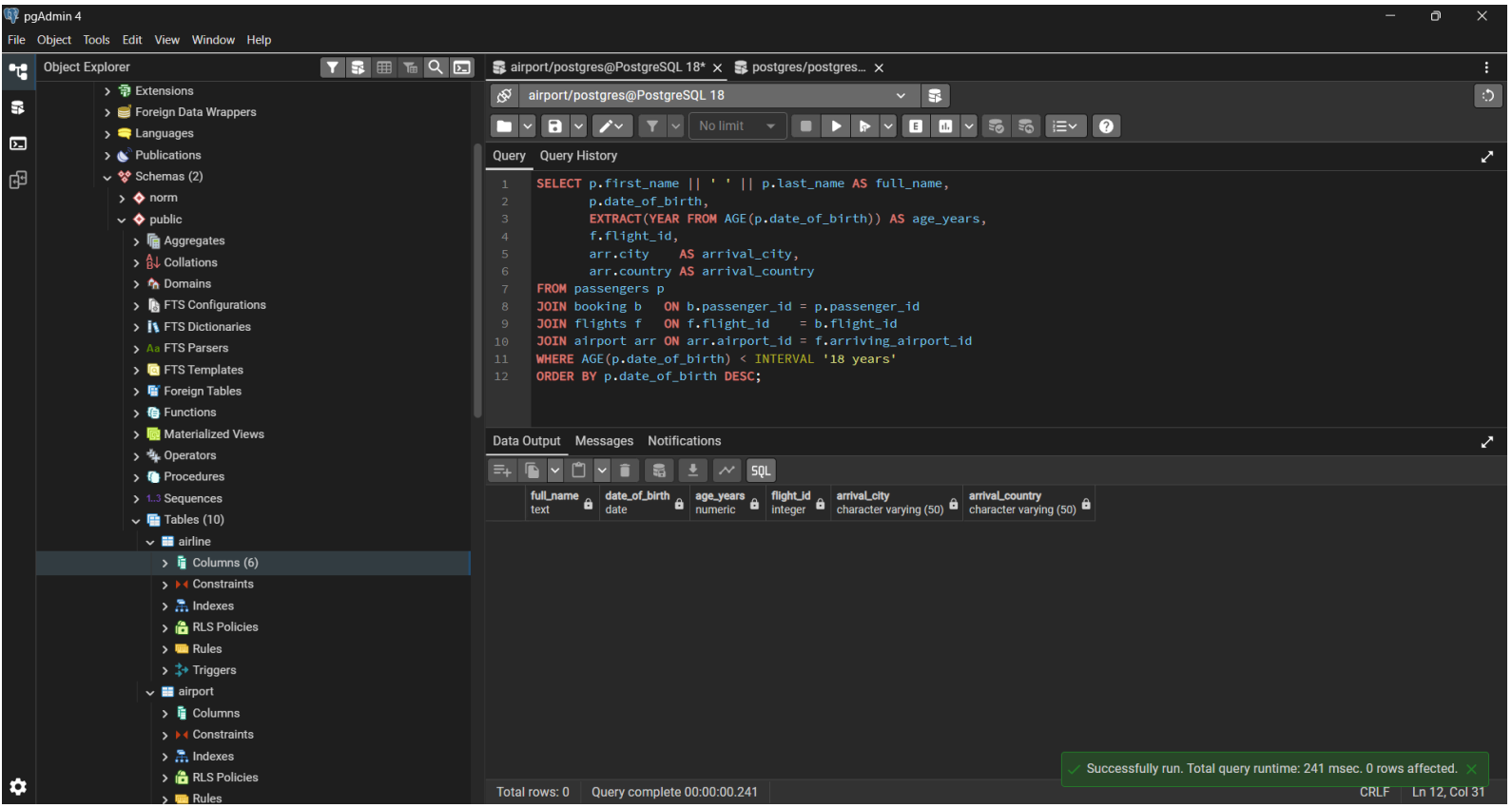
Task 5


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport/postgres@PostgreSQL 18*
postgres/postgres...
airport/postgres@PostgreSQL 18
No limit
Query Query History
SELECT f.flight_id,
count(b.booking_id) AS tickets_sold,
ROUND(AVG(b.ticket_price), 2) AS avg_price,
SUM(b.ticket_price) AS total_price,
MAX(b.ticket_price) AS max_price,
MIN(b.ticket_price) AS min_price
FROM flights f
JOIN booking b ON b.flight_id = f.flight_id
GROUP BY f.flight_id
ORDER BY f.flight_id;
Data Output Messages Notifications
Showing rows: 1 to 13 Page No: 1 of 1
flight_id [PK] integer | tickets_sold bigint | avg_price numeric | total_price numeric | max_price numeric | min_price numeric
1 11 4 46250.00 185000.00 50000.00 40000.00
2 13 1 50000.00 50000.00 50000.00 50000.00
3 14 1 50000.00 50000.00 50000.00 50000.00
4 15 2 50000.00 100000.00 50000.00 50000.00
5 16 2 23527.79 47055.57 26406.81 20648.76
6 19 1 37520.97 37520.97 37520.97 37520.97
7 23 1 50000.00 50000.00 50000.00 50000.00
8 24 2 37271.22 74542.43 39558.33 34984.10
9 25 2 24187.39 48374.77 30864.52 17510.25
10 26 1 33215.29 33215.29 33215.29 33215.29
11 28 1 33543.83 33543.83 33543.83 33543.83
12 29 2 49526.87 99053.74 50000.00 49053.74
13 30 1 50000.00 50000.00 50000.00 50000.00
Total rows: 13 Query complete 00:00:00.230 CRLF Ln 10, Col 22

Task 6

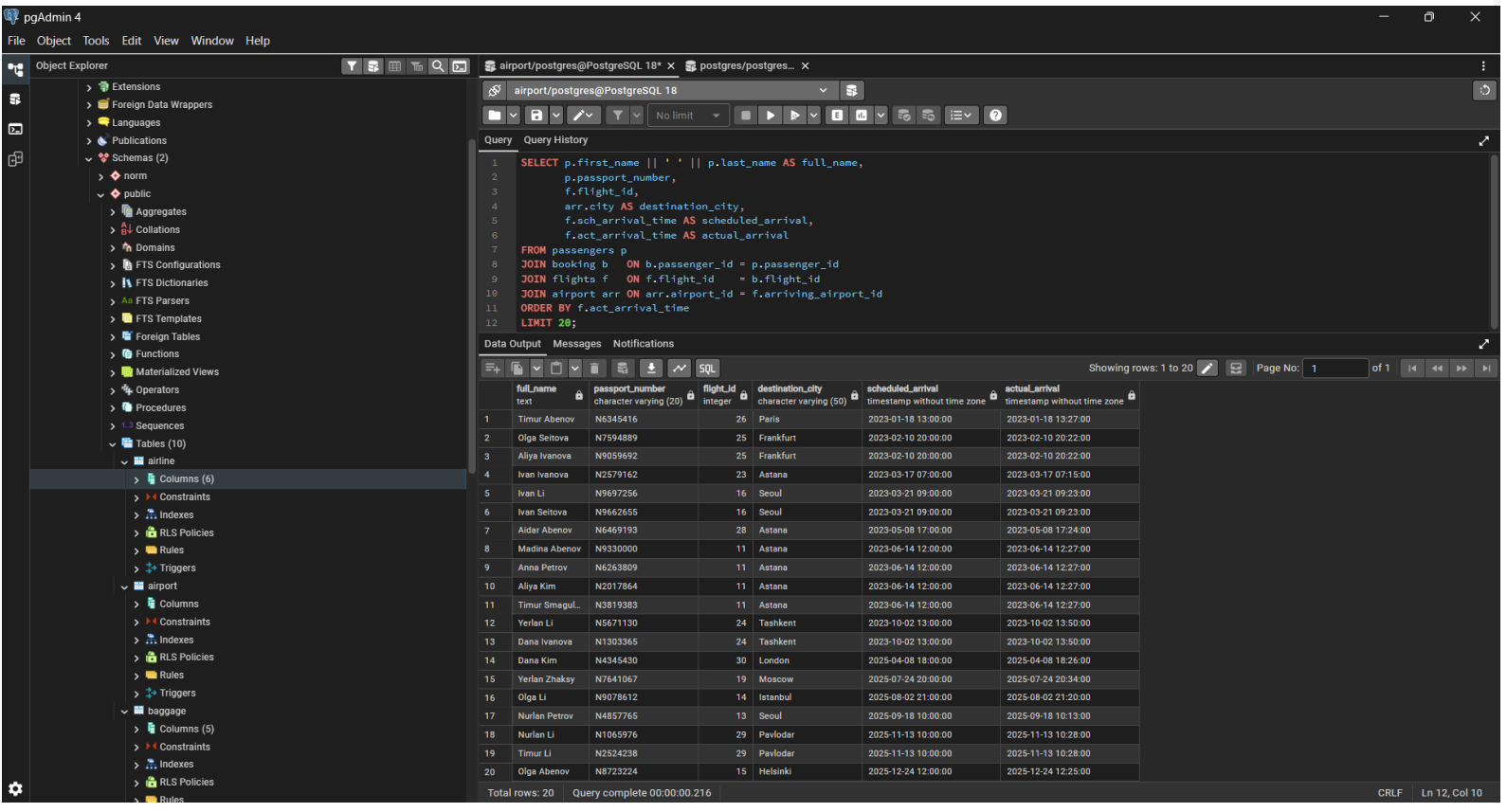

pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
airport/postgres@PostgreSQL 18*
postgres/postgres...
airport/postgres@PostgreSQL 18
No limit
Query Query History
SELECT f.flight_id,
al.airline_name,
dep.city AS from_city,
arr.airport_name AS arrival_airport,
arr.city AS to_city,
arr.country AS to_country,
f.sch_arrival_time
FROM flights f
JOIN airline al ON al.airline_id = f.airline_id
JOIN airport dep ON dep.airport_id = f.departing_airport_id
JOIN airport arr ON arr.airport_id = f.arriving_airport_id
WHERE arr.country = 'Kazakhstan'
ORDER BY f.sch_arrival_time;
Data Output Messages Notifications
Showing rows: 1 to 8 Page No: 1 of 1
flight_id integer | airline_name character varying (50) | from_city character varying (50) | arrival_airport character varying (50) | to_city character varying (50) | to_country character varying (50) | sch_arrival_time timestamp without time zone
1 23 Global Airways Tashkent Astana International Airp... Astana Kazakhstan 2023-03-17 07:00:00
2 28 KLM Seoul Astana International Airp... Astana Kazakhstan 2023-05-08 17:00:00
3 11 Aeroflot London Astana International Airp... Astana Kazakhstan 2023-06-14 12:00:00
4 31 Air Astana Almaty Pavlodar Airport Pavlodar Kazakhstan 2025-03-10 12:00:00
5 34 SCAT Pavlodar Almaty International Airp... Almaty Kazakhstan 2025-03-13 10:00:00
6 35 FlyArystan Frankfurt Almaty International Airp... Almaty Kazakhstan 2025-03-14 19:00:00
7 17 Korean Air Paris Pavlodar Airport Pavlodar Kazakhstan 2025-08-14 08:00:00
8 29 Emirates Helsinki Pavlodar Airport Pavlodar Kazakhstan 2025-11-13 10:00:00
Total rows: 8 Query complete 00:00:00.229 CRLF Ln 13, Col 29

Task 7

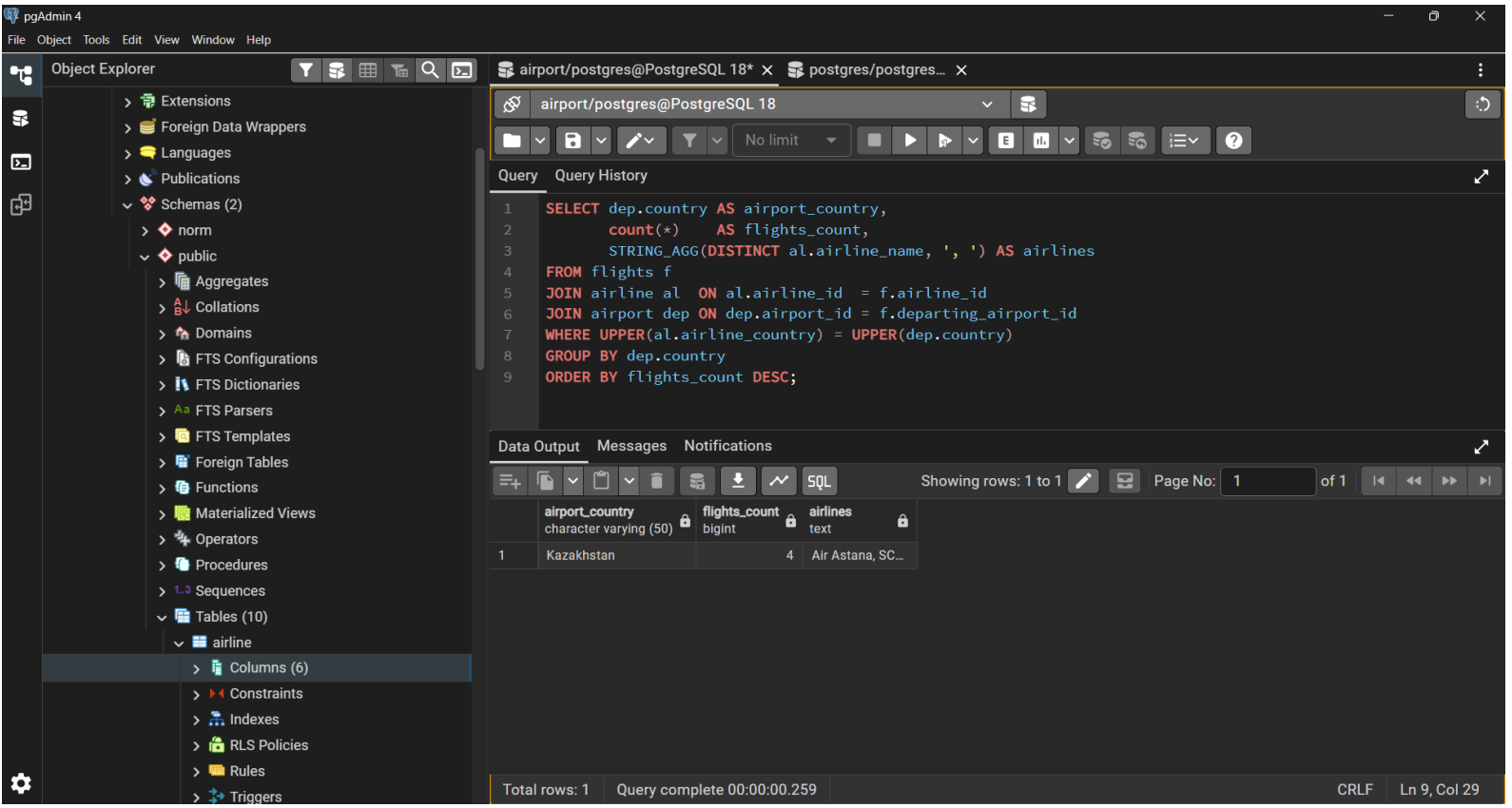


The result is empty: CHECK restrictions from Lab work n5 do not allow passengers under the age of 18 (men) and 19 (women)

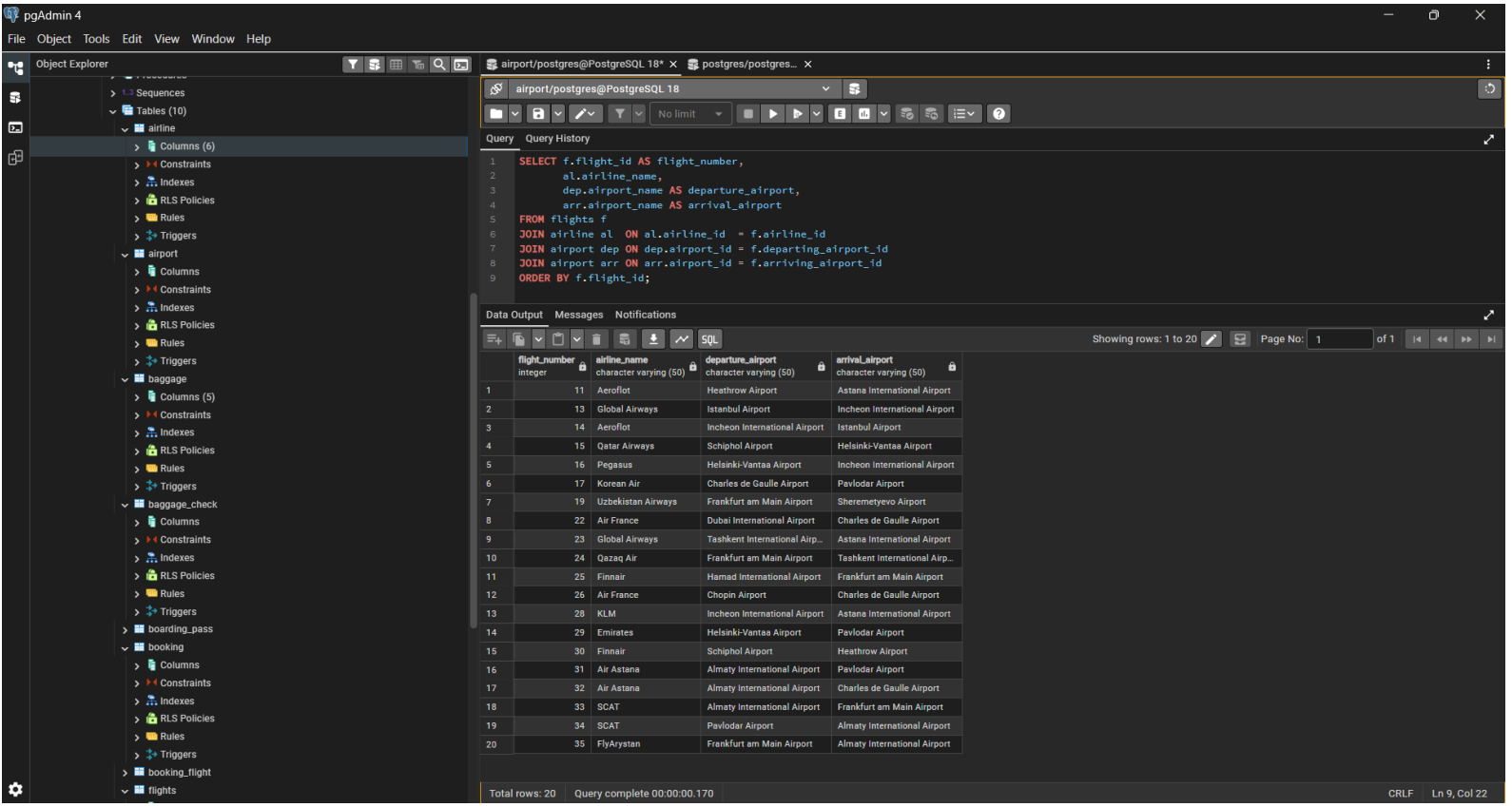
Task 8



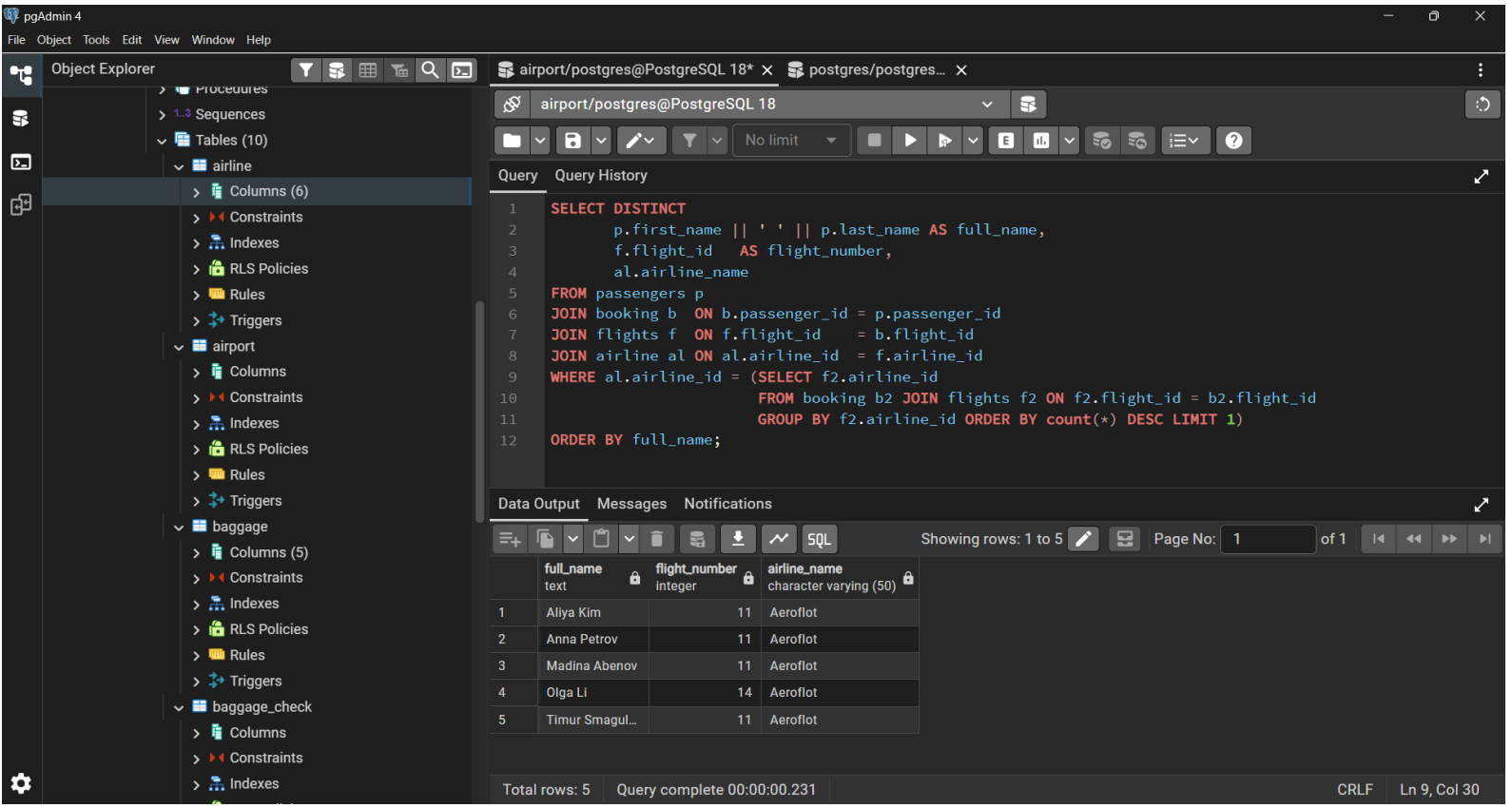
Task 9



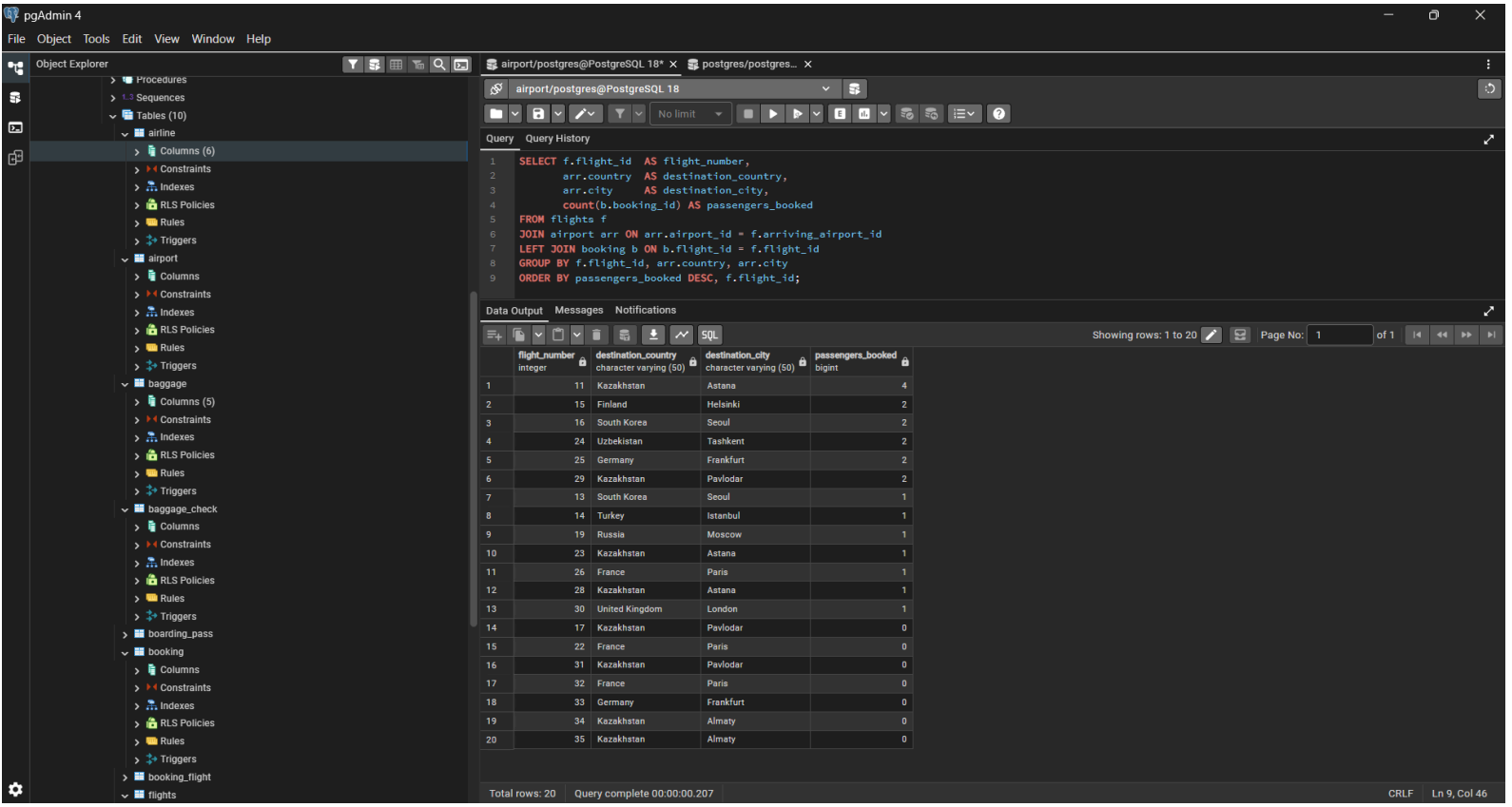
Task 10



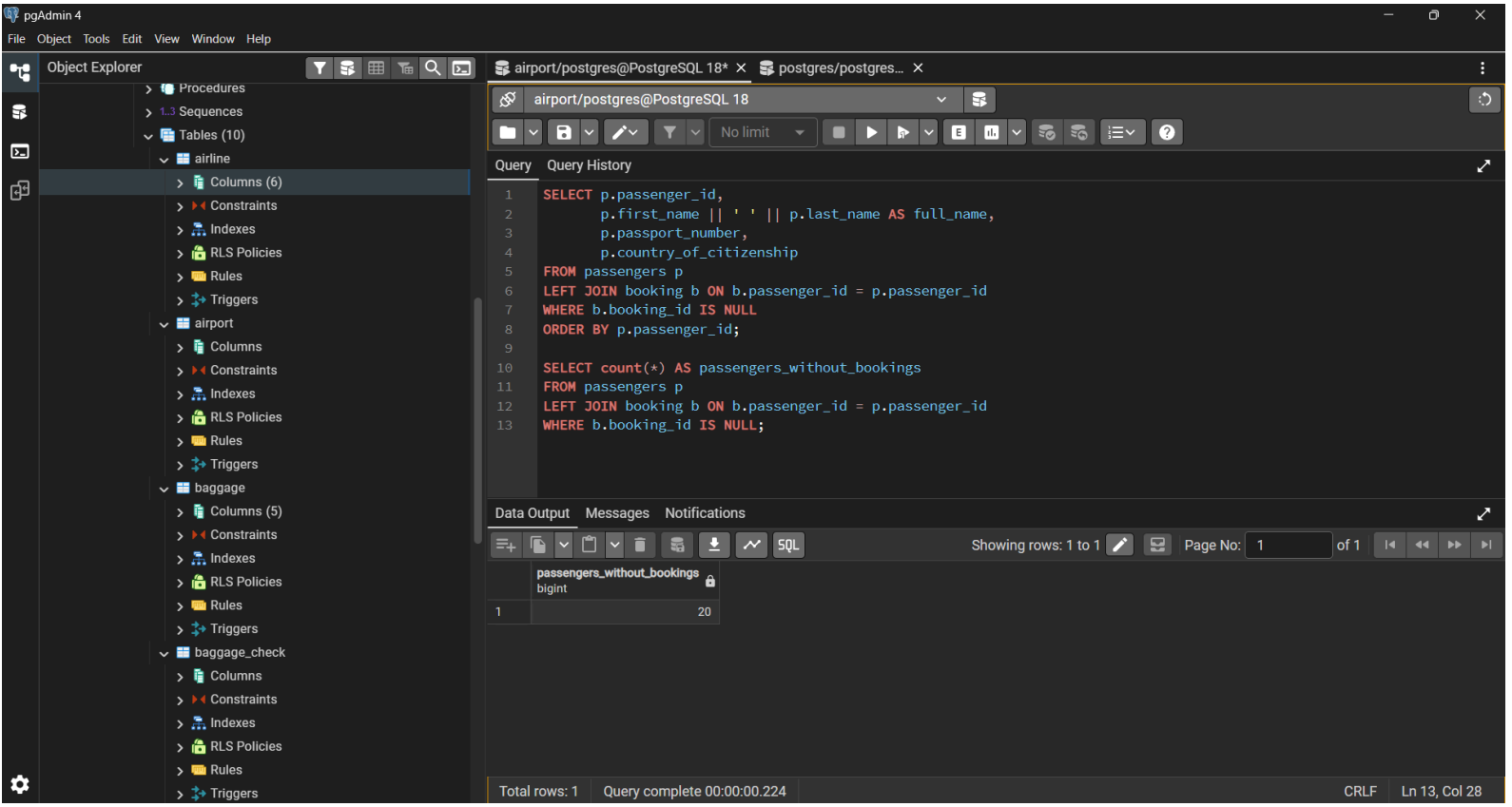
Task 11



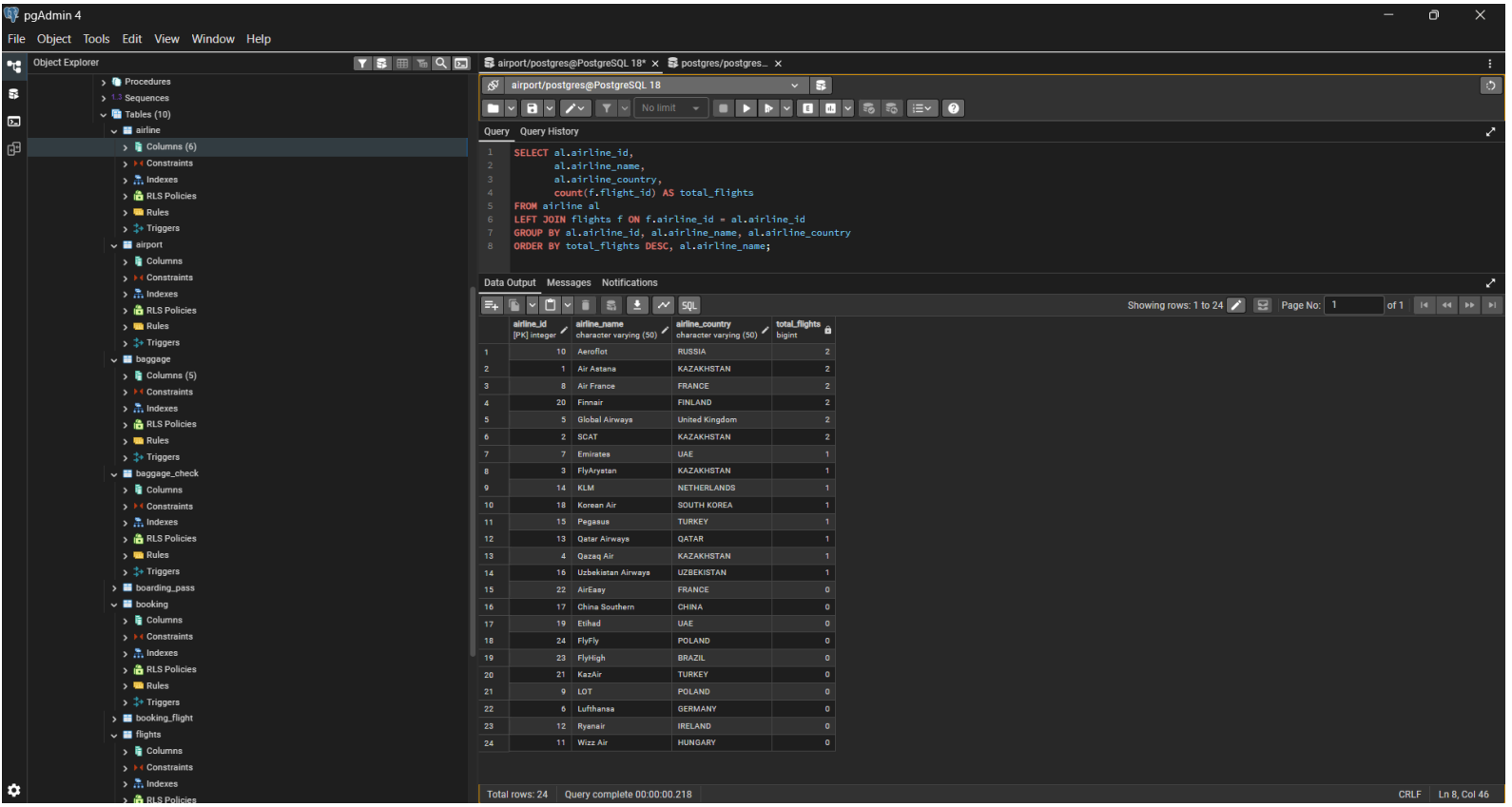
Task 12



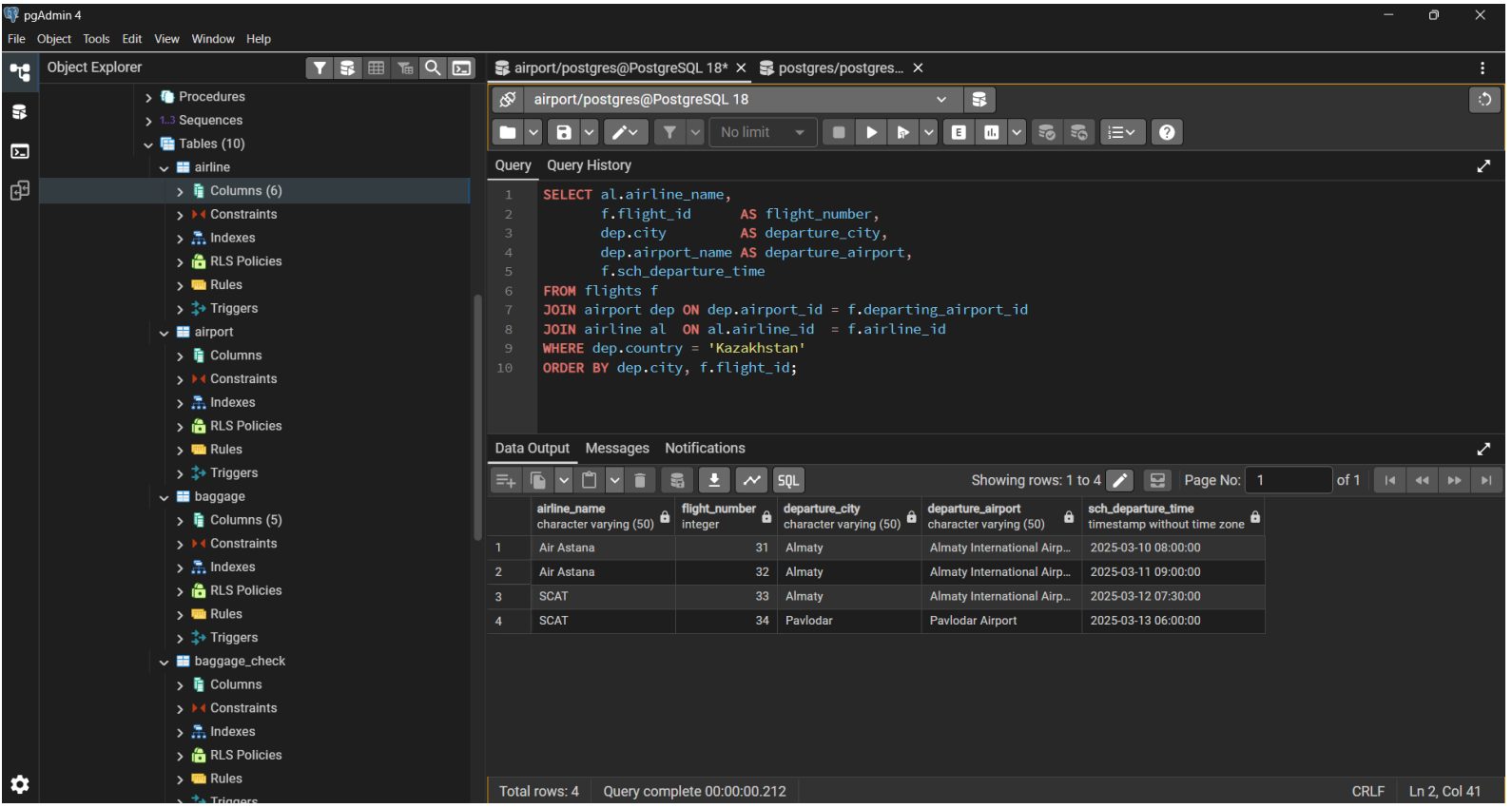
Task 13



Task 14



Task 15


pgAdmin 4
File Object Tools Edit View Window Help
Object Explorer
Procedures
Sequences
Tables (10)
airline
Columns (6)
Constraints
Indexes
RLS Policies
Rules
Triggers
airport
Columns
Constraints
Indexes
RLS Policies
Rules
Triggers
baggage
Columns (5)
Constraints
Indexes
RLS Policies
Rules
Triggers
baggage_check
Columns
Constraints
Indexes
RLS Policies
Rules
airport/postgres@PostgreSQL 18*
postgres/postgres...
airport/postgres@PostgreSQL 18
No limit
Query
Query History
SELECT al.airline_name,
f.flight_id AS flight_number,
dep.city AS departure_city,
dep.airport_name AS departure_airport,
f.sch_departure_time
FROM flights f
JOIN airport dep ON dep.airport_id = f.departing_airport_id
JOIN airline al ON al.airline_id = f.airline_id
WHERE dep.country = 'Kazakhstan'
ORDER BY dep.city, f.flight_id;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 4
Page No: 1 of 1
airline_name character varying (50)
flight_number integer
departure_city character varying (50)
departure_airport character varying (50)
sch_departure_time timestamp without time zone
1 Air Astana 31 Almaty Almaty International Airp... 2025-03-10 08:00:00
2 Air Astana 32 Almaty Almaty International Airp... 2025-03-11 09:00:00
3 SCAT 33 Almaty Almaty International Airp... 2025-03-12 07:30:00
4 SCAT 34 Pavlodar Pavlodar Airport 2025-03-13 06:00:00
Total rows: 4
Query complete 00:00:00.212
CRLF
Ln 2, Col 41